

Module 4: Electrons, waves and photons

The aim of this module is to ultimately introduce key ideas of quantum physics. Electromagnetic waves (e.g. light) have a dual nature. They exhibit both wave and particle-like behaviour. The wave–particle dual nature is also found to be characteristic of all particles (e.g. electrons).

Before any sophisticated work can be done on quantum physics, learners need to appreciate what electrons are and how they behave in electrical circuits. A basic understanding of wave properties is also required.

In this module, learners will learn about electrons, electric current, electrical circuits, wave properties, electromagnetic waves and, of course, quantum physics.

Learners have the opportunity to appreciate how scientific ideas of quantum physics developed over time (HSW7) and their validity rested on the foundations of experimental work (HSW1 and HSW2).

4.1 Charge and current

This short section introduces the ideas of charge and current. Understanding electric current is essential when dealing with electrical circuits. This section does not lend itself to practical work but to introducing

important ideas. The continuity equation ($I = Anev$) is developed using these key ideas. This section concludes with categorising all materials in terms of their ability to conduct.

4.1.1 Charge

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) electric current as rate of flow of charge; $I = \frac{\Delta Q}{\Delta t}$	
(b) the coulomb as the unit of charge	
(c) the elementary charge e equals 1.6×10^{-19} C	Learners will be expected to know that an electron has charge $-e$ and a proton a charge $+e$. HSW7
(d) net charge on a particle or an object is quantised and a multiple of e	
(e) current as the movement of electrons in metals and movement of ions in electrolytes	HSW7
(f) conventional current and electron flow	HSW7
(g) Kirchhoff's first law; conservation of charge.	